

Adult ADHD (Read more [here](#))

Genetic findings

- Genetic studies reveal a family history concordance rate of approximately 50% in first-degree relatives.
- The gene most strongly implicated in ADHD is the D₄ dopamine receptor gene DRD4.

Neuroimaging

- SPECT revealed decreased blood flow in the frontal lobes and the posterior periventricular region of the right hemisphere (Lou 1990).
- The caudate nuclei/striatum were the most consistent areas of underfunctioning
- A SPECT study of dopamine transporter density in adults with ADHD revealed density in the striatum to be elevated by about 70% (Dougherty 1999).

Biochemical risk factors for ADHD

- Food and additive allergies
- Heavy-metal and other environmental toxins
- Low-protein/high-carbohydrate diets
- Mineral imbalances
- Essential fatty acid and phospholipid deficiencies
- Amino acid deficiencies
- Thyroid disorders
- Vitamin B deficiencies

Environmental factors associated with ADHD

- Prenatal and/or perinatal risks (e.g. prematurity, low birth weight, exposure to toxins)
- Parental and family factors (e.g. child neglect, inconsistent parenting, high expressed emotion, family conflict, early institutional rearing)
- Acquired neurobiological risks (e.g. head trauma)

Research suggests that up to 66% of children will continue to have clinically significant symptoms in adulthood (Barkley 2002).

Steps in the diagnosis of ADHD in adults

- Clinical interview to assess current ADHD symptoms (within the past 6 months)
- Assess functional impairments at work, home and in relationships
- Developmental history and childhood history of ADHD symptoms

- Rule out other psychiatric disorders or establish comorbid conditions
- Family history of mental health problems (e.g. ADHD, obsessive-compulsive disorder, tics, learning difficulties, attention/concentration difficulties)
- Complete physical examination (rule out head trauma, seizures, substance misuse, hormonal problems)
- Rule out ongoing social stressors as generators of symptoms that mimic ADHD
- Neuropsychological assessments (e.g. attention, memory, intellectual functioning and academic achievement)

(Adapted from Weiss 2003)

Assessment and diagnosis of ADHD in adults

Ask the patient to indicate how often, on a 4-point scale (0, never or not at all; 1, sometimes or somewhat; 2, often or pretty much; 3, very often or very much), he or she has experienced the following symptoms in the past 6 months and whether they have persisted for at least 6 months.

Assessment

1. Inattention

- Does not pay close attention to what he or she is doing and makes careless mistakes
- Has trouble paying attention to tasks
- Has trouble following verbal instructions
- Starts things but does not finish them
- Has trouble getting organised
- Tries to avoid doing things that require a lot of concentration
- Misplaces things
- Is easily distracted by other things going on
- Is forgetful

2. Hyperactivity-impulsivity

- Fidgets with hands or feet
- Has trouble sitting still
- Feels restless and jittery
- Has trouble doing things quietly
- Is a person who is 'on the go'
- Talks too much
- Acts before thinking things through
- Gets frustrated when having to wait for things
- Interrupts other people's conversations

Diagnosis

All of the following criteria must be met for a diagnosis of ADHD:

1. six or more of the symptoms of either (a) or (b) above are rated as 'often or pretty much' or 'very often or very much' and have persisted for at least 6 months (for adults over 50, only three or more symptoms rated 'often' or 'very often' are required)
2. some symptoms of inattention or hyperactivity-impulsivity that caused impairment were present in childhood
3. impairment from the symptoms is present in two or more settings (e.g., at work, at school, at home, during leisure activities, when driving a car)
4. there is clinically significant impairment in social, academic or occupational functioning
5. the symptoms do not occur exclusively during the course of a pervasive developmental disorder, schizophrenia or other psychotic disorder and are not better accounted for by another psychiatric disorder

(After Weiss 2003, with permission)

Rating Scales

- Brown Attention Deficit Disorder Scale (Brown 1996)
- Conners Adult ADHD Rating Scales (CAARS) (Conners 1999)
- The Wender-Reimherr Adult Attention Deficit Disorder Scale (WRAADS) (Wender 1981)
- Copeland Symptom Checklist for Attention Deficit Disorders – Adult Version (Copeland 1991)
- Adult ADHD Self-Report Scale (ASRS) (Adler 2006)

Comorbidity

- Comorbidity is common, with estimates as high as 80%.
- The rate of comorbidity with personality disorders is 10–20% and with antisocial behaviour 18–28% (Barkley 1998).
- The comorbidity for alcohol misuse is 32–52%; for other types of substance misuse, including marijuana and cocaine, it is 8–32%
- Dysthymia in particular is common in adults with ADHD.
- Impulsivity in ADHD is associated with psychiatric comorbidity and may present both directly and indirectly with forensic sequelae

Treatment

Medication	Formulations	Starting daily dosage	Target daily dosage
Methylphenidate	5, 10, 20 mg tablets; 20 mg slow-release tablets	5–10 mg, 3–4 times daily	40–60 mg
	Controlled-release tablets: 18, 36 mg tablets	18 mg, once daily in the morning	36–54 mg
Clonidine	25, 100, 150 µg	25 µg	150–300 µg
Pemoline	18.75, 37.5, 75 mg tablets; 37.5 mg chewable tablets	37.5 mg	75 mg
Atomoxetine	10, 18, 25, 40, 60 mg capsules	0.5mg/kg	1.4 mg/kg
Dexamphetamine	5, 10, 15 mg slow-release capsules; 5 mg per 5 ml (elixir)	5–10 mg, 2–3 times daily	20–45 mg
Methamphetamine	5 mg tablets; 5, 10, 15 mg long-acting tablets	5–10 mg	20–45 mg
Mixture of amphetamine and dexamphetamine	5, 10, 20 mg tablets	5–10 mg	20–45 mg
Imipramine	10, 25, 50 mg tablets	10–25mg	100–150 mg
Nortriptyline	10, 25, 50, 75 mg capsules	10–25mg	75–100 mg
Bupropion	75, 100 mg tablets	37.5 mg, twice daily	150–300 mg

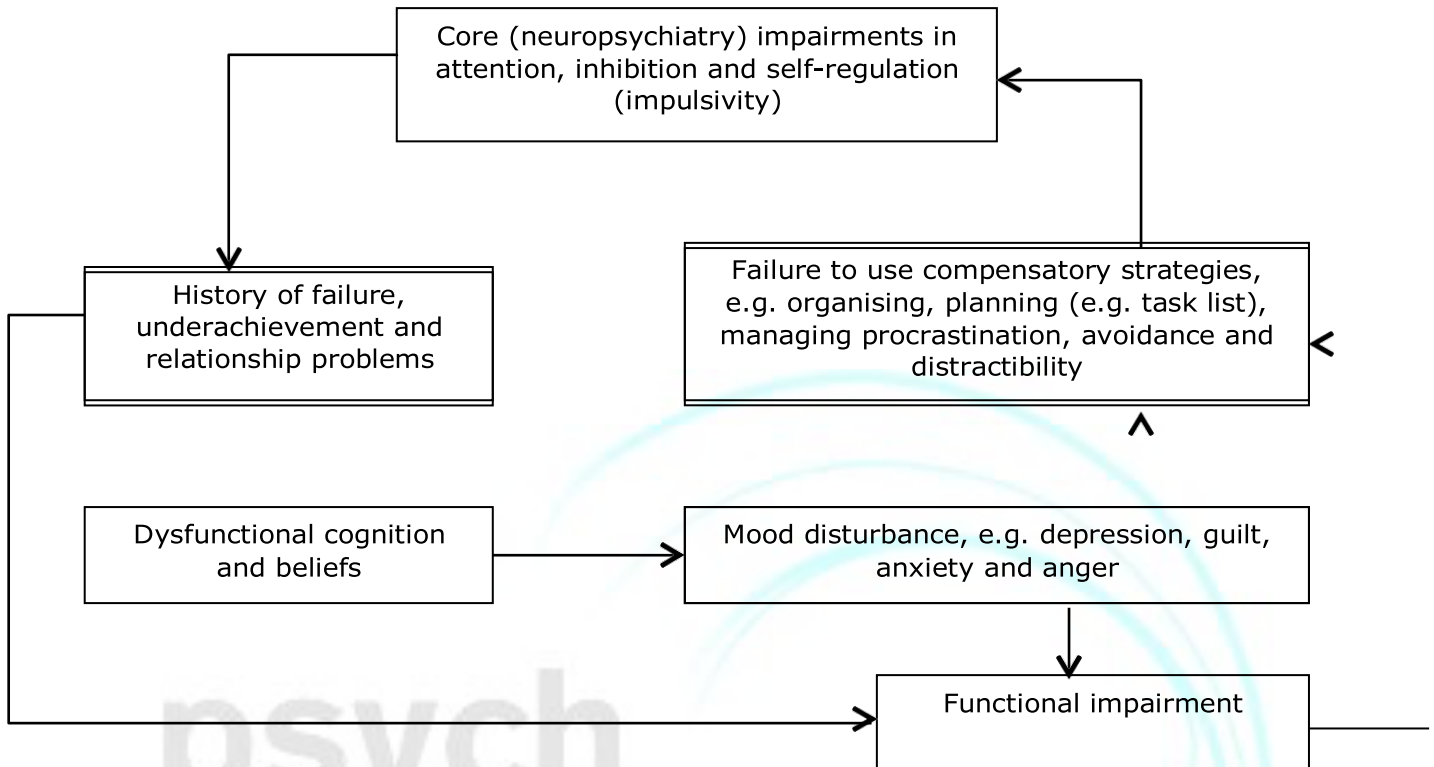
First-line treatment

- The first line of treatment is with stimulant agents such as methylphenidate or amphetamine salts.
- Methylphenidate dosing requires careful titration from lowest possible dose to achieve best control of symptoms.
- Extended-release methylphenidate tablets should not be crushed, divided or chewed.
- Dexamphetamine can be started at a dose of 10 mg/day, and increased by 10 mg/day weekly up to a maximum of 45 mg/day. About two-thirds of adults with ADHD who are given these medications show significant improvement in their symptoms (Gadow 2001).
- Atomoxetine: It is a highly specific nor-adrenaline reuptake inhibitor with minimal affinity for serotonin and dopamine transporters and neuronal receptors.

Second-line treatment

- Second-line treatment includes imipramine, clonidine and, occasionally, moclobemide.

A cognitive-behavioural model of impairment in adult ADHD (after Safren 2004, with permission).



Cognitive-behavioural therapy for ADHD

First module

- This provides training and structure for enhanced organisational and planning skills. The first three sessions involve learning new skills and the fourth session is for review and consolidation. The module also includes training in problem-solving skills.

Second module

- This targets distractibility.

Third module

- Targets the elimination of dysfunctional thoughts that increase negative affect and that enhance avoidance of work-related topics.

Optional modules

- Application of skills to cope with procrastination, anger, communication and relationship problems.
- Marital/individual counselling and self-help groups are often valuable adjuncts to pharmacotherapy and skills training.

Environmental interventions

- Changes in physical setting can improve individuals' learning and functioning.

ADD and Substance Use Disorder

Relationship between ADHD and substance use disorder

- ADHD greatly elevates risk for substance use and misuse
- High novelty seeking trait, self-medication for ADHD symptoms and the presence of conduct disorder and bipolar disorder are risk factors for substance misuse in adult ADHD
- Patients with ADHD and substance use disorder tend to commence early and experiment more freely with substance misuse
- Growing up with ADHD confers a greater risk for using alcohol and tobacco, whereas clinic-referred adults with ADHD seem more likely to use marijuana, cocaine and LSD
- Substance use may be considered as a form of self-medication for patients with ADHD

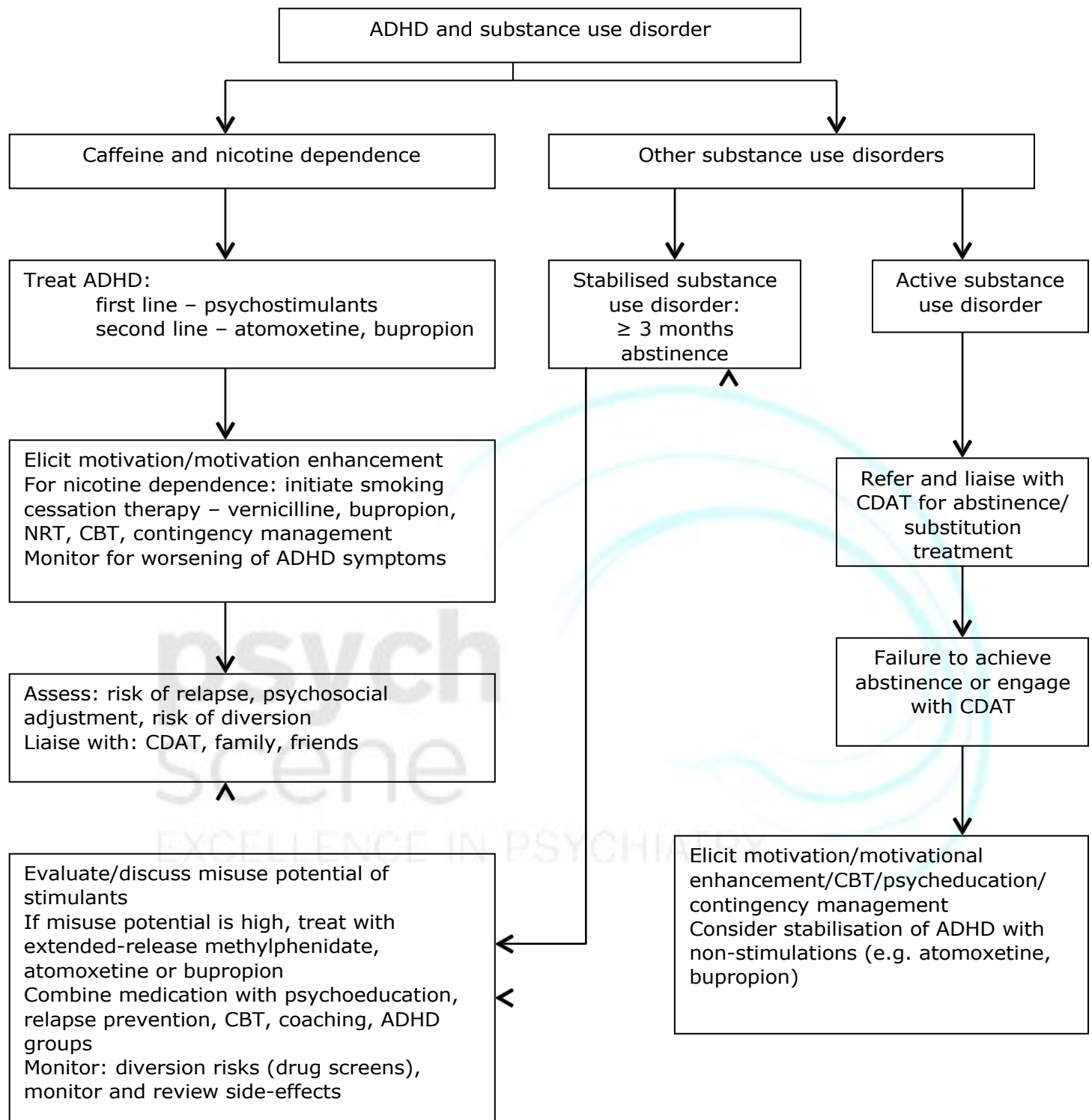
Assessment of ADHD and substance use disorder

- Patients presenting with substance use disorder should be **screened for presence of ADHD** (Screening instruments)
- Assessment of these individuals should include **current and childhood history of ADHD symptoms, detailed history of current and past substance use, previous treatments, and psychiatric, family and forensic history**
- At least 1 month of abstinence is useful for accurate and reliable assessment for ADHD symptoms
- It is imperative to watch for **signs of possible misuse**, such as **missed appointments**, and **signs of possible diversion**, such as **repeated requests for higher doses** and a pattern of 'lost' prescriptions
- It is important to exclude high-risk situations (e.g. comorbid antisocial personality disorder, strong forensic history and family member or peers with substance use disorder)

Treatment

- In a meta-analysis using data from six studies (five prospective longitudinal studies and one retrospective study) Wilens et al (2003) reported a 1.9-fold reduction in risk of substance use disorders in individuals with ADHD who had taken stimulant medications compared with similar individuals who had not taken such medications.
- Several longitudinal studies (Molina 2007; Barkley 2008; Biederman 2008) found no significant effect of stimulant treatment, age or duration of stimulant treatment on the development of substance misuse.
- Newer extended-release stimulants and pro-drug formulations minimise the risk of misuse and diversion of drugs (Spencer 2006).

Treatment algorithm for the management of attention deficit-hyperactivity disorder (ADHD) comorbid with substance use disorder. CBT, cognitive-behavioural therapy; CDAT, community drugs and alcohol team; NRT, nicotine replacement therapy.



References

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Nicotine Addiction and Management in Mental Illness

Summary of effective smoking cessation medication

- NRT is effective for smoking cessation
- Combining different forms of NRT enhances cessation rates
- Pre-loading with NRT may increase cessation rates
- Use of NRT alongside a reduction in cigarette consumption may also be useful as a first step towards cessation for heavier dependent smokers
- Bupropion and nortriptyline are effective but SSRIs and anxiolytics do not improve cessation rates
- Bupropion reduces withdrawal symptoms as well as weight gain
- Varenicline is also effective and may be more effective than bupropion
- Six forms of NRT: gum, patch, inhalator, nasal spray, tablet and lozenge.
- Combining a nicotine patch with a rapid-delivery form of NRT is more effective than a single type of nicotine replacement (Stead et al, 2008).
- There is some evidence that using the nicotine patch for a short period before the quit attempt results in higher cessation rates (Stead et al, 2008).
- A systematic review (Hughes et al, 2007) showed that the **antidepressant bupropion doubles smoking cessation rates** (odds ratio OR = 1.94, 95% CI 1.72–2.19) as does nortriptyline (OR = 2.34, 95% CI 1.61–3.41).
- **Varenicline is a nicotine receptor partial agonist** and is thought to act both as an agonist maintaining moderate levels of dopamine to counteract withdrawal symptoms and as an antagonist by reducing smoking satisfaction.
- Side-effects of varenicline include gastrointestinal disturbance such as nausea, dry mouth, taste disturbance, headache, drowsiness, dizziness, sleep disorders and abnormal dreams. Case reports describe varenicline's induction of a manic episode in a patient with bipolar affective disorder (Kohen & Kremen, 2007) and its exacerbation of the symptoms of schizophrenia (Freedman, 2007). Depression and suicidal ideation have been reported
- There is insufficient data regarding whether naltrexone is helpful for smoking cessation (David et al, 2006).

Summary of medication and psychological treatment for those with depression

- NRT can double cessation rates and lower self-reported depression in depressed smokers
- One study showed that extended use of bupropion for relapse prevention was effective for smokers with and without a history of major depression
- Psychological and lifestyle strategies assisted in mood regulation over and above the standard smoking cessation treatments for smokers with a depression history
- Structured psychological approaches to smoking cessation added to the benefits of antidepressants and NRT particularly for smokers with high nicotine dependence and repeated episodes of major depression

- CBT doubled abstinence rates in a trial of smokers with recurrent depression
- Increased cessation rates for those being treated for depression occurred through addition of counselling and computerised motivational feedback

Smoking cessation in schizophrenia

Medication

- Nicotine replacement is effective, although less than in the general population
- Bupropion contributed to reduced smoking and did not worsen clinical symptoms although relapse rate was high after treatment discontinuation
- Combination NRT and bupropion treatment is effective in abstinence and reduction

Combined medication and psychological therapy

- Specialised group therapy plus NRT is effective, and atypical antipsychotics improve abstinence with such treatment

Harm reduction

- Harm reduction approaches include NRT and NRT plus bupropion to reduce the level of smoking, which may improve the chance of future cessation

Pharmacokinetic Interactions

- Smoking affects the metabolism of various psychiatric medications by inducing enzymes in the cytochrome P450 (CYP) system, potentially lowering serum levels of medication by as much as 50% (Wilhelm et al, 2004).
- Medication metabolised by CYP1A2 includes diazepam, haloperidol (partial), olanzapine (partial), clozapine, mirtazapine (partial) and tricyclic antidepressants (Bazire, 2003).
- Smoking can significantly lower serum levels of such medication. Haslemo et al (2006) estimate that a daily consumption of 7–12 cigarettes is probably sufficient for maximum induction of clozapine and olanzapine metabolism, and recommend a 50% lower starting dose in non-smokers to avoid side-effects.
- Clozapine plasma levels have been found to be 40% lower in smokers than in non-smokers (Seppala et al, 1999). Clozapine toxicity has also been observed during early tobacco abstinence in patients being treated with clozapine (Zullino et al, 2002).

Reference

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Benzodiazepine Dependence

Benzodiazepines enhance the effects of the neurotransmitter gamma-aminobutyric acid (GABA). They do so through action at benzodiazepine receptors. Alleviation of fear is a result of GABA-induced inhibition of neuronal transmission that occurs in the amygdala, located in the brain stem, and thought to mediate brain circuits involved in the appraisal of threat and experience of fear.

Commonly used benzodiazepines

Short-acting

- Alprazolam
- Lorazepam

Intermediate-acting

- Oxazepam

Long-acting

- Diazepam
- Ethyl loflazepate
- Clorazepate
- Chlordiazepoxide
- Clonazepam

When to consider benzodiazepines for anxiety disorders

Clinical situations in which benzodiazepines may be considered as the initial treatment for anxiety disorders (with or without an antidepressant) include:

- panic disorder (and to a lesser extent, generalised anxiety disorder and social anxiety disorder)
- need for rapid symptomatic relief (e.g. for severe anxiety and frequent panic attacks)
- history of severe side-effects of the previously administered antidepressant(s), especially if these side-effects resulted in premature cessation of the medication
- absence of current or past substance misuse

When to consider antidepressants for anxiety disorders

Clinical situations in which antidepressants are more likely to be considered as the initial treatment for anxiety disorders include:

- post-traumatic stress disorder, obsessive-compulsive disorder
- generalised anxiety disorder, social anxiety disorder (possibly in combination with a benzodiazepine)
- presence of a co-occurring depressive disorder or history of depression

- predominance of cognitive aspects of pathological anxiety (e.g. pathological worry)
- current or past alcohol or other substance misuse/dependence, concern about benzodiazepine dependence and misuse
- NICE guidelines state that 'benzodiazepines are associated with a less good outcome in the long term and should not be prescribed for the treatment of individuals with panic disorder' and also that 'benzodiazepines should not usually be used for more than 2 to 4 weeks for the treatment of generalised anxiety disorder' (NICE 2014).
- 'If there is no history of drug dependence, and positive indicative "lifestyle" factors are present, a conscious decision to continue benzodiazepine treatment may be more reasonable than the alternatives, provided the patient periodically attempts to slowly reduce the dosage at regular intervals and tries to stop altogether when or if possible' (**Baldwin 2013).
- 'There are clinical circumstances in which longer-term prescription of benzodiazepines might be considered desirable because the alternatives are probably worse than the continued use [...] In rare instances longer-term prescriptions [...] may be seen as a form of harm reduction in patients who would otherwise consume illicit benzodiazepines' (**Baldwin 2013).

Common misconceptions about long-term benzodiazepine treatment of anxiety disorders

Misconception	Status/evidence
Tolerance to the anxiolytic effect	Vast majority of patients with anxiety disorders do not show a tendency to increase the dose and do not exhibit loss of the therapeutic benefit
Frequent misuse	Benzodiazepine misuse rarely occurs in patients with anxiety disorders who do not have current or past substance misuse problems
Frequent addiction	Substance addiction and benzodiazepine dependence are different phenomena, with little overlap
Dangerous withdrawal symptoms	Withdrawal symptoms may occur in the course of benzodiazepine taper and are very likely to occur on abrupt discontinuation, but they are rarely dangerous; the rise of withdrawal should be minimised through gradual and individualised taper
Debilitating side-effects	Benzodiazepines are generally well tolerated and there is no evidence that they cause organ toxicity; their side-effects, especially sedation and impairment of motor coordination, should be closely monitored

Characteristics of benzodiazepine withdrawal syndrome

Non-specific symptoms

- Increased anxiety
- Physical symptoms of anxiety
- Restlessness
- Sleep disturbance
- Nausea, stomach cramps
- Numbness or tingling sensations
- Unsteady gait
- Irritability
- Feeling depressed
- Feeling weak
- Tiredness

Specific symptoms

- Hypersensitivity to light, sound, smell or taste
- Feeling of things moving, as if being on a boat
- Tinnitus
- Distorted body image
- Muscle cramps and pains, myoclonic jerks
- Flu-like symptoms
- Suspiciousness
- Feeling confused
- Depersonalisation, derealisation
- Seizures (rarely)

Principles in discontinuing Benzodiazepines

- Depending on the dose, the type of benzodiazepine (short-acting, intermediate-or long-acting), duration of its use, and the patient's overall condition and personality characteristics, the taper can be completed in weeks or over a period of many months.
- Patients need to feel ready for benzodiazepine discontinuation in order for it to be successful. In other words, they need to indicate that they feel capable of coping with the anxiety without having to continuously rely on the medication. **If patients resist cessation, it is a bad idea to antagonise them by insisting that they stop 'no matter what'.**
- Refusing to continue prescribing a benzodiazepine or using other coercive measures is only likely to lead to more difficulties. It is also important to be flexible about the duration of taper, for example, by not setting rigid time limits.
- The taper should ideally proceed under the physician's supervision because they are best placed to suggest the pace of decreasing the dose, to monitor the patient's progress and to provide support.
- The last is especially important if withdrawal symptoms occur during taper or if patients become ambivalent about continuing the taper.
- Cognitive-behavioural therapy (CBT) may also be effective in facilitating benzodiazepine discontinuation (Otto 1993; Spiegel 1994).
- Both rebound and relapse may call for recommencement of the original pharmacotherapy, consideration of a different pharmacological approach, or a combination of pharmacological and psychological interventions. Adding CBT to ongoing pharmacotherapy may decrease the risk of relapse

Measures to ensure safe, effective use of benzodiazepines for anxiety disorders

- Before starting a tapering plan, inform the patient about the risks of withdrawal. Abrupt reductions from high-dose benzodiazepines can result in seizures, psychotic reactions, and agitation.
- Understanding the tapering regimen enhances compliance and outcomes. Stress the importance of careful adherence and provide close psychosocial monitoring and fail-safe means for patient contact if someone is experiencing difficulties. Supportive psychotherapy improves the prognosis.
- On a clinical basis, additional, adjunctive, symptomatic, or other medications may be required for safe illness management.
- Managing comorbid medical conditions and psychopathologies—including addressing other substances of abuse—is important.
- Tapering one or more substances at a time—even nicotine—is not advised. Refer patients to a self-help group or substance abuse rehabilitation program.
- Slow tapering is safer and better tolerated than more abrupt techniques.
- If the patient experiences overt clinical signs of withdrawal, such as tachycardia or other hyperadrenergia during dosage reduction, maintain the previous dosage until the next tapering date.
- For persons who take a short-acting benzodiazepine—eg, alprazolam or lorazepam—convert the dosage into an equivalent dosage of a long-acting benzodiazepine—eg, diazepam.
- Metabolized slowly, with a long half-life, diazepam allows a consistent, slow decline in concentration while tapering the dosage. This helps avoid severe withdrawal. For patients who have been taking alprazolam or clonazepam, 1 mg, the equivalent diazepam dosage would be 20 mg; for temazepam, 30 mg, the diazepam dosage would be 15 mg; for lorazepam, 1 mg, oxazepam, 20 mg, or chlordiazepoxide, 25 mg, the diazepam dosage would be 10 mg.^{1,2} Prescribe the to-be-tapered benzodiazepine at five-sixths of that dose and prescribe one-sixth of the diazepam amount daily.
- Proceed with tapering every 1 to 2 weeks by a one-sixth dose reduction of the tapered medication and a one-sixth increase in diazepam. Continue until diazepam is used alone and well-tolerated.
- Once the patient is taking only diazepam, decrease the dosage by 2 mg every 2 weeks until the patient is doing well on a relatively small dosage of diazepam. Subsequent diazepam reductions are at 1 mg less every 1 to 2 weeks, until the patient is able to completely discontinue the medication.
- Continue monitoring until clinical stability is achieved or otherwise indicated. Be aware that some people might switch to other substances of abuse.

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Cannabis Misuse in Mental Illness

Why do people with severe mental illness use cannabis?

- The main reasons given for cannabis use are to reduce boredom, to socialise and to enjoy the positive mood that intoxication provides
- Nearly half of patients use cannabis to get relief from dysphoria and agitation and to sleep better; a minority use it to reduce their suspiciousness
- Self-medication has been proposed as a reason for cannabis use, but this is yet to be supported by research evidence
- The cannabidiol in cannabis has strong anti-anxiety and antipsychotic effects, but street cannabis now contains very small amounts of cannabidiol

None of the models proposed to explain the interaction between severe mental illness and substance use disorder has a satisfactory evidence base, and it is thought that a multiple risk factor model is needed

Elements of a comprehensive history of cannabis use

- Age at first cannabis use
- Duration of cannabis use
- Amount of use over a set period of time
- Type and strength of cannabis used over the period of time
- Preferred method of use
- Reasons for use, including what the patient gains or loses from it
- Subjective experiences and whether these have changed over time
- Effects on mental and physical health, social life, medication, treatment adherence and finances, as experienced by the patient

Other substance use (same information gathering for each substance used)

Assessment Tools

1. Addiction Severity Index
2. Stages of Change Readiness and Treatment Eagerness Scale (SOCRATES), which is a 19-item self-report measure (Miller & Tonigan, 1996).
3. Substance Abuse Treatment Scale (SATS)
4. Substance Use Scale for Psychosis (Spencer et al, 2002) and the Readiness to Change Questionnaire (James et al, 2004)
5. The Cannabis Experiences Questionnaire is a recently developed instrument for exploring subjective experiences of cannabis use (Barkus et al, 2006). The questionnaire has three sub-scales: Pleasurable Experiences, Psychosis-Like Experiences and After-Effects.

Management

Psychological Intervention

1. Motivational Interviewing

The aim of motivational interviewing and motivational enhancement therapy for substance misuse is to strengthen the patient's motivation to change by increasing their awareness of the impact of substance use and encouraging them to take responsibility for their behaviour (Miller 2002)

The five stages of change in relation to substance misuse

- **Pre-contemplation:** continuous use with no interest in quitting in past 6 months
- **Contemplation:** continuous use with ambivalent interest in quitting
- **Preparation:** continuous use with interest in quitting in the next 30 days
- **Action:** active attempt to quit
- **Maintenance:** abstinent for more than 3 months but less than 5 years (After Prochaska 1992)

2. Motivational Interviewing plus Cognitive Behavioral Therapy

3. Insight adherence abstinence in first episode schizophrenia.

To prevent psychotic relapses, the first year of treatment targets the unique characteristics of such individuals, focusing on building adherence to medication regimens, abstinence from substance use and insight into schizophrenia. The model provides supportive, cognitive-behavioural, behavioural and motivational therapies, skills building and psychoeducation

Pharmacological

Clozapine

- A review of substance misuse by people with treatment-resistant schizophrenia revealed that clozapine produced similar improvements in symptoms and psychosocial functioning levels of both users and non-users, and that a history of substance misuse did not appear to have a negative influence on response to clozapine (Buckley 1994).
- **In a prospective study involving alcohol- and cannabis-using patients with schizophrenia and schizoaffective disorder, those receiving clozapine had significantly better remission rates for substance use than those on typical antipsychotics (Drake 2000).**
- A 10-year follow-up report from the same group showed that individuals taking clozapine were less likely to relapse into substance use over the subsequent year than those treated with other antipsychotics (Brunette 2006b)
- In a retrospective survey of 58 clozapine clinic patients with schizophrenia or schizoaffective disorder and substance use, a significant reduction in

cannabis, alcohol and cocaine use was recorded in those who continued to take clozapine, compared with those who discontinued it (Zimmet 2000).

- There are some suggestions that fluoxetine and buspirone are associated with reduced cannabis use, and nefazadone may help alleviate withdrawal symptoms (McRae et al, 2003).
- In the future, there may be therapeutic use of cannabinoid receptor antagonists, which have been shown to block the acute physiological and psychological effects of smoking cannabis (Huestis et al, 2001).

Components of a relapse prevention programme

- Identification of high-risk situations and triggers for cravings
- How to say no to buying or using cannabis; role-play may be used
- Development of skills to cope with cravings, e.g. self-talk, distracting activities; clients might keep a diary to document how cravings are managed
- Identifying alternative pleasurable activities
- Learning to cope with lapses; clients may develop a personal emergency plan to identify steps for dealing with relapse
- Management of low mood and anger using problem-solving skills and relaxation training; identification and modification of negative thoughts
- Creation of a personal self-help sheet documenting the benefits of not using cannabis, risks of using and high-risk situations

(Marlatt & Gordon, 1985)

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Medically Unexplained Symptoms

(Somatic Symptom Disorder, Somatisation, Conversion Disorder, Hypochondriasis)

Risk Factors

1. Intergenerational transmission

The children of parents who present with medically unexplained symptoms are at greater risk of developing such symptoms than are the offspring of parents with organic medical conditions (Levy et al, 2001; Craig et al, 2002).

2. Abuse in childhood

Sexual and physical abuse and neglect in childhood are linked to a greater risk of both psychological and physical problems in adulthood.

3. Environmental stressors

In adults, environmental stressors such as divorce or bereavement can precipitate both psychological reactions and somatic syndromes.

4. Amplification of Bodily Sensations (Servan-Schreiber AAFP 2000)

5. Psychodynamic Issues- The Identified Patient

When a family system is under stress, identifying one person as a patient may provide a focus that stabilizes the family system and alleviates feelings of anxiety within the family. Members know how to interact with each other in the context of the illness. The patterns of behavior may become recurrent, and family rules about how each member should act are formed. The patterns may become dysfunctional when one member takes on the role of being weak and defective.

6. Sick Role

The somatizing patient seems to seek the sick role, which affords relief from stressful or impossible interpersonal expectations ("primary gain") and, in most societies, provides attention, caring and sometimes even monetary reward ("secondary gain").

7. Illness Beliefs

Patients with medically unexplained symptoms have more negative illness perceptions and poorer health status than patients without such symptoms (Frostholtm et al, 2007).

Illness beliefs in relation to symptoms

Belief	Symptom
Normalising	I'm tired because I've been working too hard
Somatic	I'm tired because I've got a virus
Psychological	I'm tired because I've been stressed

Treatment Principles

1. Discussing Diagnosis with Empathy and acknowledge disability

Physicians should be comfortable making a statement such as: "The results of my examination and of the tests we conducted show that you do not have a life-threatening illness. However, you do have a serious and impairing medical condition, which I see often but which is not completely understood. Although no treatment is available that can cure it completely, there are a number of interventions that can help you deal with the symptoms better than you have so far."

Do

- Use one designated physician.
- Schedule frequent, brief, regular visits not contingent on new complaints.
- Allow "sick role;" focus on function rather than symptoms.
- Explore psychosocial issues.
- Prescribe benign treatments and enjoyment time.

Problems in Management

Several common problems can derail the course of treatment.

1. Setting Unrealistic Goals
2. Presence of Co-morbid Medical and Psychiatric Conditions

Patients with somatization disorder commonly have coexisting depression (up to 60 percent), anxiety disorders such as panic or obsessive-compulsive disorder (up to 50 percent), personality disorders (up to 60 percent) or a substance abuse disorder.

3. Requests for diagnostic studies
4. Requests for opiate analgesics
5. Frequent Telephone calls
6. Repeat attendance to Emergency Department
7. Negative Transference and Countertransference reactions

CARE-MD Approach

(Managing Somatization Disorder, *Robert McCarron, DO*)

Cognitive Behavioral Therapy	Follow the cognitive behavioral therapy treatment plan developed by the therapist and patient
Assess	Rule out potential general medical causes for the somatic complaints
Regular visits	Short frequent visits with focused exams Discuss recent stressors and healthy coping strategies Patient should agree to stop overutilization of medical care (e.g. frequent emergency room visits or excessive calls and pages to the primary care provider, etc.)
Empathy	"Become the patient" for a brief time During visits, spend more time listening to the patient rather jumping to a diagnostic test Acknowledge patient's reported discomfort
Med-psych interface	Help the patient self-discover the connection between physical complaints and emotional stressors ("the mind-body" connection) Avoid comments like, "your symptoms are all psychological" or "there is nothing wrong with you medically"
Do no harm	Avoid unnecessary diagnostic procedures Minimize consults to medical specialties Once a reasonable diagnostic workup is negative, feel comfortable with a somatoform-type diagnosis and initiate treatment

Cognitive Behavioural Therapy

Patients should also be instructed to use a daily dysfunctional thought record (DTR) to monitor their depressive or anxious emotions and associated negative thoughts. In collaboration with the therapist, psychiatrist can learn to use basic cognitive behavioral techniques and quickly review the DTR during office visits.

The reattribution model

Stage 1 Feeling understood

Stage 2 Broadening the agenda

Stage 3 Making the link

Stage 4 Negotiating treatments

Lifestyle Recommendations

e.g. psychoeducational advice about dealing with marital conflict or more effective ways to handle relationships with children. Family system interventions that reinforce healthy family alliances may also help.

Physical exercise is important, even in patients who believe themselves to be physically impaired. It prevents deconditioning, enhances self-esteem and provides an opportunity for patients to take a break from oppressive duties or unpleasant situations. A minimum of three 20-minute exercise sessions per week is desirable.

Treat Co-Morbidities**Reference**

1. Servan-Schreiber, David, Gary Tabas, and N. Randall Kolb. "Somatizing patients: part II. Practical management." *American family physician* 61.5 (2000): 1423-1436.



Personality Disorder

An enduring pattern of inner experience and behavior that deviates markedly from the expectations of the individual's culture, is pervasive and inflexible, has an onset in adolescence or early adulthood, is stable over time, and leads to distress or impairment [American Psychiatric Association 1994: p. 629]."

DSM-IV classification of personality disorder (Axis II)

Cluster A: odd or eccentric disorders

- Paranoid personality disorder: irrational suspicions and mistrust of others
- Schizoid personality disorder: lack of interest in social relationships, seeing no point in sharing time with others, anhedonia, introspection
- Schizotypal personality disorder: odd behaviour or thinking

Cluster B: dramatic, emotional or erratic disorders

- **Antisocial personality disorder:** pervasive disregard for the law and the rights of others
- **Borderline personality disorder:** extreme 'black and white' thinking, instability in relationships, self-image, identity and behaviour, often leading to self-harm and impulsivity
- **Histrionic personality disorder:** pervasive attention-seeking behaviour, including inappropriately seductive behaviour and shallow or exaggerated emotions
- **Narcissistic personality disorder:** pervasive pattern of grandiosity, need for admiration, and a lack of empathy

Cluster C: anxious or fearful disorders

- **Avoidant personality disorder:** social inhibition, feelings of inadequacy, extreme sensitivity to negative evaluation and avoidance of social interaction
- **Dependent personality disorder:** pervasive psychological dependence on other people
- **Obsessive–compulsive personality disorder:** rigid conformity to rules, moral codes and excessive orderliness

(American Psychiatric Association 1994)

Major components of clinical features in personality disorder

- An intrapersonal component:** marked individual dysregulation of arousal, impulse and affect systems in response to stress; individuals typically present as either hyperaroused or hypoaroused in unpredictable ways
- An interpersonal component:** dysfunctional interpersonal attachment patterns that reduce healthy functioning and are a further source of stress; these may take the form of getting too close or being detached and uninvolved

- c. **A social component:** dysfunctions in social behaviours which bring individuals with personality disorders into conflict with others and sometimes into contact with statutory agencies such as mental health or criminal justice systems

Symptom subtypes in personality disorders

- **Arousal:** Over- or under arousal, often self-medicated with drugs or alcohol; enhanced tendency to dissociate in cluster B; emotional indifference
- **Affect:** Anger; suspicion; fearfulness; detachment; coldness or restricted emotion; resentfulness; anxiety about new activities; irritability; low mood; intense distress; feelings of rage and fear of abandonment; fear that others are threatening or attacking them; rapidly shifting emotions
- **Cognition:** Poor reality testing in terms of dissociation; brief psychotic episodes; odd beliefs and magical thinking; cognitive distortions; preoccupations and ruminations in cluster C
- **Somatic self-identity:** Somatic disorders, ranging from preoccupations with the somatic self to attacks on the physical body in the form of self-harm
- **Psychological self-identity:** Disorders of sense of self; lack of sense of agency; exaggerated sense of self-importance; belief that the self is in danger from others; difficulty in distinguishing self from others

Pathogenesis

1. The impact of external events on neurobiological development and gene–environment interaction

Being raised in a hostile or abusive environment is posited to increase the risk of developing personality disorder because of the direct effect of chronic stress on the developing cytoarchitecture of the autonomic nervous system, limbic system, amygdala and the right orbitofrontal cortex (Schore 2001). The basic task of child care consists in responding sensitively to episodes of interactive signals produced by the autonomic nervous system in both infant and carer. These episodes emerge when the infant reaches about 2 months of age, and they are highly arousing, affect-laden and short interpersonal events that expose the infant to high levels of cognitive and social information (Feldman 1999).

As the infant grows, it is the memory of the relationship, rather than the particular caregiver, that becomes the (accessory) affect regulator, allowing attachment to develop with others on the basis of this template. A good-quality affect regulatory system, based on secure bonding between carer and infant, leads to optimal maturation of the right hemisphere of the brain at a critical period during the first 2–3 years of life (Schore 2003). The right orbitofrontal cortex (ROFC) acts as a regulator and modulator both of amygdala responses to fear and distress and of autonomic nervous system response (LeDoux 1996; National Scientific Council on the Developing Child 2011). Any measurable damage to cytoarchitecture of the developing ROFC results in failure to develop a top-down regulatory system (Taylor 1997; Schore 2003).

Among boys, it has been shown that genetically determined low monoamine

oxidase A (MAOA) activity, leading to dysregulation of the 5-HT system, moderates the association between childhood maltreatment and later vulnerability to the effects of environmental stress, thus causing mental health problems (Kim-Cohen 2006).

2. Theories that address social attachment over time

It has been suggested that insecure attachment should be seen as a dimension of personality disorder (Livesley 1998). The pathology of borderline personality disorder is associated with a subtype of insecure attachment called preoccupied or enmeshed attachment, whereby the individual is highly ambivalent about those to whom they are attached (either in a passive or angry way, or both). Individuals who display antisocial violent behaviour towards others are more likely to exhibit a dismissing attachment pattern, in which weakness or vulnerability are denied and attachment to others is seen as unnecessary or contemptible (Pfafflin 2003; van IJzendoorn 2009).

3. Theories of personality disorder as persistence of childhood thinking/feeling patterns

Other theorists have emphasised the importance of thoughts, feelings and beliefs as organised 'defences' against distress, and suggested that personality disorder may be best understood as a collection of immature defences (Vaillant 1993, 1994; Bond 2004; Cramer 2006).

Psychological defences are those personality traits, cognitions and beliefs that help an individual regulate their own sense of distress. Studies of people with personality disorder indicate that they almost exclusively use immature defences, and do not use mature defences (Vaillant 1994).

Assessment

Principles in Assessment

1. History Taking

- risk of harm to self and others
- the presence of other mental health difficulties
- the complexity of a person's personality difficulties
- the level of burden and/or distress placed on other family members or agencies.

2. Risk Assessment: Factors to be considered during assessment

- Demographic factors
- Current social situation
- Current presentation
- Psychosocial stressors

- Previous history of violence and self-harm
 - Previous response to treatment/supervision
 - Level of social support
 - Anger
 - Impulsivity
 - Substance misuse
 - Presence or absence of mental illness
- (Doyle 2006)

3. Presentation

It is important to carry out an assessment over several interviews. This allows the interviewer to be more confident that the patient's presentation reflects personality traits rather than their current mental state. It is also important to bear in mind that fluctuation in presentation may itself be a characteristic of personality disorder (e.g. emotional lability in borderline personality disorder).

4. Clinical Interview

The interviewer has the opportunity to reflect not only on the content of the response but also on the emotional expression and any non-verbal communication. The patient's response to the assessor and the feelings evoked in the assessor also inform the assessor's understanding of the patient's interpersonal functioning and difficulties.

It is important to allow them the opportunity to identify which part of their interpersonal functioning causes them most distress and what they wish to change or modify. A joint understanding of a patient's treatment goals helps to build a positive therapeutic alliance. It is important to enquire about high-risk behaviour directed against the self and others, as this affects treatment and management.

Assess maladaptive behaviour, its effect on the individual and others, attitudes and relationships with others, and social functioning in all areas of the person's life over a prolonged period of time. The interviewer must assess both the individual's current functioning and their normal functioning throughout their life

Other Sources of Information

Supplement interview with other sources of information: Records, Family members, staff etc.

Structured Assessments

1. Structured personality disorder assessment instruments

Structured categorical (diagnostic) assessments

Observer-rated structured interview

- International Personality Disorder Examination (Loranger 1994)
- Diagnostic Interview for DSM–IV Personality Disorders (Zanarini 1996)
- Structured Interview for DSM–IV Personality Disorders (Pfohl 1997)
- Structured Clinical Interview for DSM–IV Axis I Disorders (First 1997)
- Personality Disorder Interview–IV (Widiger 1995)

Self-rated questionnaire

- Personality Diagnostic Questionnaire (Hyler 1994)

Structured interview – other sources

- Standardised Assessment of Personality (Mann 1981)
- Personality Assessment Schedule (Tyrer 1979)

Structured dimensional assessments

Observer-rated structured interview

- Schedule for Normal and Abnormal Personality (Clark 1990)

Self-rated questionnaire

- Personality Assessment Inventory (Morey 1991)
- Minnesota Multiphasic Personality Inventory–II (Butcher 1989)
- Millon Clinical Multiaxial Inventory–III (Millon 1997)
- Eysenck Inventory Questionnaire (Eysenck 1975)
- NEO Five-Factor Inventory (McCrae 1992)

Unstructured assessments

Interview based

- Clinical interview
- Psychodynamic formulation

Other

- Rorschach test (Rorschach 1964)
- Thematic Apperception Test (Morgan 1935)

Minnesota Multiphasic Personality Inventory–II

The Minnesota Multiphasic Personality Inventory–II (Butcher 1989) is a self-report measure of global psychopathology consisting of 567 true/false items giving information about symptoms and interpersonal relationships.

2. Comorbidity

Patients with one diagnosed personality disorder often have further personality disorders and other dysfunctional personality traits and mental health problems. The presence of comorbidity should be explored in the psychiatric history, and additional assessment instruments should be administered to check for further personality disorders

3. Severity

The type of personality disorder diagnosed and an understanding of its impact on functioning gives an indication of the severity of the disorder.

4. Treatability

- **Intellectual functioning**
- **Psychological Mindedness**
- **Readiness for treatment:** the presence of characteristics (states or dispositions) within either the client or the therapeutic situation, which are likely to promote engagement in therapy and which, thereby, promote therapeutic change. (Howells and Day, 2007). Readiness is a function of both internal (patient) and external (context) factors. **Internal factors include cognition, affect, volition, behaviour and identity. External factors include circumstances, location, opportunity, resources, support and available treatment.** Internal factors that suggest a greater level of readiness for treatment include: a positive appraisal of the treatment intervention offered; ability to trust others; a capacity to express emotions and reflect on emotional states; moderate (but not overwhelming) distress; experiencing guilt rather than shame; and a belief that change is possible.

5. Transference and Countertransference: (Moore, E. (2012). Personality disorder: its impact on staff and the role of supervision. *Advances in psychiatric treatment*, 18(1), 44-55.)

Service users may

- compare various members of nursing and multidisciplinary teams and their decisions
- provide minimal information to a member of staff to see how they cope

- make high numbers of complaints
- question others with the same therapist/nurse about actions that person took previously
- refuse to follow recommendations thereby emphasising staff vulnerability, not their own
- seek to relate to staff by adopting their language, style and affectations
- single staff out for special attention/comment, gifts or favours, criticism or assault
- play different professionals off against one another, exploiting normal inter-professional rivalries
- flirt with or reject staff

Staffing Issues that arise with Personality Disorders

1. Tensions between personal and professional identities

2. **The pull to punish:** Bullying, intimidation and assaults on patients by staff have been associated with circumstances in which there are no safeguards on imbalances of power, and the majority operate in collusion with a custodial regime (Blom-Cooper 1992; Fallon 1999).

3. Displacement, 'acting out' of distress and Splitting

4. **Complex and damaging repetitions:** Staff actions/interpersonal styles often trigger such conflict re-enactments through the process of 'projective identification' (whereby 'bad' feelings, impulses and beliefs are projected onto staff, who experience the patient's emotions as their own). For example, a frightened patient may consider his named nurse to be persecutory and be unusually hostile towards her (due to fear), while the named nurse experiences her patient as frightening. Staff may find themselves subjected to the hatred and rejection that patients were exposed to in earlier circumstances (Neeld 2009).
5. **Boundary Violations:** Slippages (behaviour that falls into the grey area between staying within or crossing boundaries) preceding violation are known to include overfamiliarity, inappropriate favouritism and the exchange of promises/gifts (Webb 1997; Walker 1999).

Management-led strategies to promote health and recovery in personality disorder services

- Retain professional roles: always keep the duty of care in focus
- Select motivated staff
- Provide training on personality disorders, boundary setting and maintenance
- Attend to changes in staff attitude/interest/morale
- Promote an 'empathic balance' between trust and control: share decisions (rules agreed by peers are more likely to be accepted than those imposed by parental figures)
- Establish a culture of reflective enquiry: 'Can we safely explore together what

might be happening here?’

- Include appropriate use of humour
- Respond swiftly but calmly to rule-breaking; operate with ‘firmness without retribution’; do not retaliate; do not over-react
- Provide opportunities for detained persons (and staff) to ‘let off steam’
- Communicate the learning points from serious incidents/set-backs specifically and directly to all staff

Strategies in Other Personality Disorders

Fagin, L. (2004). Management of personality disorders in acute in-patient settings. Part 2: Less-common personality disorders. *Advances in Psychiatric Treatment*, 10(2), 100-106.

1. Schizoid and Schizotypal Personality Disorder

Interactive issues in the management of patients with schizoid personality disorders

- Accept silence as a powerful form of communication
- Be prepared to feel rejected and distanced
- Be careful of persistent probing
- Accept the patient’s rhythm and pace of change
- Validation by other patients often carries more weight than validation by staff

2. Antisocial Personality Disorder: (after Lion, 1999)

- Remain sceptical, particularly during assessments
- Do not deny or normalise dangerousness: a charismatic patient can lure staff into a forgiving or permissive response to unacceptable behaviour
- Be aware of polarities in feeling: a patient’s assertiveness or even violence can arouse secret awe and admiration one moment, but disgust and punishment the next
- Be aware of the risk of sexual seduction by patients
- Ensure that less experienced staff are supervised by a more experienced clinician

3. Histrionic Personality

- First, the initial assessment should be used as an opportunity to challenge the patient’s cognitive style.
- Second, erotic transference must be effectively managed. Eroticised feelings towards staff can sometimes be very insistent and pervasive, and in an in-patient unit nursing staff are particularly vulnerable to overt or covert advances.

4. Avoidant Personality Disorder

- offer an empathic response to their embarrassment, coupled with encouragement to face the feared situation;
- consider prescription of a selective serotonin reuptake inhibitor;
- use psychoeducational approaches to combat anxiety.

References

1. Adshead, G., & Sarkar, J. (2012). *The nature of personality disorder. Advances in Psychiatric Treatment, 18(3), 162-172.*
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Transference and Countertransference

The therapeutic alliance

There are three parts to the therapeutic relationship: **the therapeutic alliance, the transference and the countertransference**

- The **therapeutic alliance** is the rational (implicit) contract between doctor and patient
- The contract may be straightforward with mutual cooperation
- The contract may be complicated by a covert agenda: the patient's unconscious and unspoken wishes and needs (the transference)

Transference

- Transference is unconscious
- It is at least partly inappropriate to the present
- It is the transferring of a relationship, not a person
- Only an aspect of a relationship, not the entire relationship, is transferred

Projection and transference

- Transference involves the projection of a mental representation of previous experience on to the present
- Other people are treated as though they are playing the complementary role needed for the projected relationship
- There are subtle (unconscious) behavioural “nudges” to take on these feelings and behaviours

Managing transference

- **Recognising the importance of the relationship to the patient**
- **Reliability:** Establishing a relationship with a dependent patient is a necessary part of treatment and the inevitable transference must be sensitively handled. If the doctor or keyworker is unreliable, this will not lessen the transference but will complicate it. Making and keeping appointments is professional and is calming for the patient.
- **Maintaining professional boundaries** and clear limits in treatment
- **Interpretation**, but only when the patient can understand and use it

Countertransference

- Includes the feelings evoked in the doctor by the patient's transference projections
- These can be a useful guide to the patient's expectations of relationships
- They are easier to identify if they are not congruent with the doctor's personality and expectation of his or her role

- Awareness of the transference-countertransference relationship allows reflection and thoughtful response rather than unthinking reaction from the doctor

Patients who do not get better (From Hughes and Kerr 2000)

*"Patients who appear to have bypassed the rational therapeutic contract and who resist recovery over a long period of time despite getting good conventional treatment may be a particular source of frustration. **Some frankly want a simple contact relationship with the doctor and have no interest in treatment or cure.** If the doctor gets angry and rejecting, it is likely to increase the patient's anxiety and intensify his or her demands or cause them to go to a colleague. Most doctors or teams resolve this problem with limited but reliable non-interventionist contact."*

"Ms F was a 50-year-old woman who had many depressive episodes and somatic complaints throughout her life. She was a frequent attender at her general practitioner's (GP's) surgery. The GP found that a 15-minute monthly appointment kept her relatively well, and that her demands did not escalate."

***Some patients have a more destructive agenda with a wish to engage the doctor in a therapeutic endeavour with the ultimate aim of proving that he or she, the patient, is untreatable.** This interaction requires a helpful person who keeps trying. Such patients may leave a string of failed therapists in their wake.*

"Mr G is a man of 30 who lives with his parents and despite high intelligence works in a low-paid clerical job. He has had many treatments for depression and 'inability to sort [his] life out'. Whatever is advised, and whatever interpretations made, he returns to the next session to explain why any change has been impossible. At the end of his most recent failed therapy, he says sympathetically, 'I don't want you to feel bad doctor, I've defeated five therapists before you'."

Reaction and reflection (From Hughes and Kerr 2000)

"Reaction might be called therapist acting-out. It happens when we either play the role unconsciously given to us by the patient or when we are aware of not being seen as we are and respond with anxiety or anger. Much of the time we have to tolerate not understanding what is going on without panicking. We should not be provoked into precipitate and premature action simply to reduce our own anxiety."

"In the weekly ward round, the consultant sees that a patient with personality disorder has been an in-patient for six weeks. He feels that the patient is exploiting the service. He says angrily that the patient is getting dependent and must be discharged by the end of the week. (See also 'Transference in reverse' above.)"

"Reflection demands a reasonable level of awareness of one's own thoughts and feelings, and a sound grasp of whether these deviate from good professional behaviour. Good practice includes:

- *questioning attitude towards one's own feelings and motives*
- *recognition that we all have 'blind spots'*
- *understanding that staff are affected by patients*

- *understanding that patients are affected by staff behavior*
- *recognition that patients often have strong feelings towards staff."*

Dealing with countertransference

- Reflection: see above
- Using the team to clarify what a difficult patient projects into the treatment relationship – often, several members of the group can contribute to an understanding of the patient's transference to the team or to individuals (Kerr, 1999)
- Using a specialist psychotherapist to help understand what the patient is unconsciously communicating in his or her behaviour
- Undergoing personal therapy to become more aware of one's own unconscious needs and fears.

References

1. Hughes, P., & Kerr, I. (2000). Transference and countertransference in communication between doctor and patient. *Advances in Psychiatric Treatment*, 6(1), 57-64.



Clozapine Treatment and Issues

- Clozapine is the only antipsychotic drug licensed for treatment-resistant schizophrenia.
- Kane et al (1988) introduced rigorous criteria to define treatment resistance, emphasising persistence of positive symptoms despite receiving 'adequate treatment'. This involved at least three treatment periods of 6 weeks or more with first-generation antipsychotics (FGAs) within the past 5 years at doses equivalent to 1000 mg/day of chlorpromazine. Inadequate response was measured in terms of failure to improve by at least 20% in total Brief Psychiatric Rating Scale (BPRS) score.
- Revised criteria: A trial of two antipsychotics including one second-generation antipsychotic (SGA) for 4–6 weeks at a dose equivalent to 400–600 mg of chlorpromazine is an 'adequate trial' (Conley 2001). When evaluating 'adequate response', effect on quality of life is now considered as important as symptom reduction (Lewis 2006).
- Treatment resistance affects up to 33% of people with schizophrenia (Essali 2009), although studies using broader definitions of the term suggest that prevalence may be as high as 60% (Essock 1996).
- A recent 11-year follow-up study of over 66 000 patients with schizophrenia in Finland concluded that antipsychotic treatment was associated with reduced mortality compared with no antipsychotic use and that clozapine was the most effective antipsychotic in this regard (Tiihonen 2009).
- Clozapine is the recommended treatment for treatment-resistant schizophrenia because of robust evidence for its superior efficacy compared with FGAs and other SGAs (Leucht 2009)
- The current consensus is that up to 60% of patients with treatment-resistant schizophrenia will show improvement with clozapine (Meltzer 1997). However, it can take from 6 to 12 months for this to occur (Meltzer 1989). Two to three years of treatment may be required to stabilise symptoms enough to enable patients to return to work and avoid readmissions (Lindstrom 1988; Woodall 2004).
- Clozapine has also been established as a cost-effective treatment for treatment-resistant schizophrenia. Studies have reported significant resource savings mainly due to reduced frequency of hospital readmissions (Aitchison 1997; Hayhurst 2002).
- In a study of 120 patients conducted in four hospitals in south-east London, Taylor et al (2003) found a mean of 9.2 episodes of antipsychotic prescription before clozapine was initiated. The mean delay in using clozapine was 5 years. High-dose SGAs and polypharmacy are often used instead of clozapine (Fayek 2003).
- A telephone survey of 100 psychiatrists in Denmark found that many were reluctant to prescribe clozapine, had little experience in its use and believed that patients treated with clozapine were less satisfied than those treated with other antipsychotics (Nielsen 2010). A survey of consultant psychiatrists working in adolescent units in the UK reported similar concerns.
- Patients who complete a year of clozapine treatment have a higher rate of continuation to 2 or more years than those who discontinue treatment at an

early stage (Munro 1999). This may be due to the length of time it takes for beneficial effects to become evident.

- Younger age at initiation of clozapine has consistently been shown to be associated with longer treatment maintenance, whereas the probability of discontinuation increases with advancing age (Munro 1999; Whiskey 2003).

Issues in management of patient with clozapine

1. Adherence: *"Refraining from instituting clozapine because of an expectation that the patient will not adhere to treatment once back in the community is not an excuse to make no attempt to try it, particularly if the patient is currently detained for treatment under the Mental Health Act."* Mortimer APT, 2011. There is some evidence that patients maintain adherence with clozapine better than with other antipsychotic treatments (Wheeler 2009; Kelly 2010). Furthermore, there is limited recent evidence that co-prescription of depot antipsychotic treatment provides a 'safety net' for patients who repeatedly fail to take their clozapine properly (Kim 2010).

2. Suicide and other risks

3. Problems with Clozapine

Problem	Solution
Clozapine level suboptimal	Adjust dose above 'therapeutic threshold', check interactions
Positive symptoms with adequate clozapine level	Augmentation options
Negative symptoms	Augmentation options, check levels if using an SSRI
Cognitive impairment, poor function	Rehabilitation, social support, wait and see
Depression	Characterise and treat accordingly; if 'awakenings', close personal support, decelerate expectations
Neutropenia	Consider risks and benefits of initiation/rechallenge, seek consultant haematology support, consider granulocyte stimulating factor, 'rescue' contingency or prophylaxis, Lithium prescription
Eosinophilia/thrombocytopenia	Stop until resolved
Fever	Antipyretic, monitor for signs of infection, check WCC, ECG, creatine phosphokinase
Tachycardia	ECG, monitor for cardiac symptoms, try a beta blocker
Myoclonic jerks	Drop dose if possible, small dose of clonazepam at night

Clozapine dose above 600 mg daily or level >0.60	Sodium valproate seizure prophylaxis
Oversedation	Drop dose, split (small dose in the morning, larger at night) or give dose all at night, stop other sedating medication
Constipation	Dietary advice, laxatives, drop dose if possible
Urinary incontinence	Assess, use anticholinergic smooth muscle relaxant if appropriate, drop dose if possible
Sialorrhoea	Hyoscine, drop dose if possible
Weight gain, hyperglycaemia, dyslipidaemia	Dietary advice, exercise, metformin, add aripiprazole
Risk of unintentional pregnancy	Full discussion with patient, contraceptive advice or referral if appropriate
Discontinuation – Rebound Psychosis	Sudden discontinuation of clozapine is very often followed by rebound psychosis, which can be severe and very difficult to treat. Rebound psychosis can be complicated by cholinergic rebound, which can cause nausea, vomiting, diarrhoea, headache, sweating, restlessness and agitation. This at least can be treated easily with an anticholinergic drug. Even so, clozapine discontinuation should take place gradually, unless this cannot be avoided (for example, if clozapine has to be stopped abruptly because there is neutropenia).

References

1. Mistry, H., & Osborn, D. (2011). Underuse of clozapine in treatment-resistant schizophrenia. *Advances in Psychiatric Treatment*, 17(4), 250-255.
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HIV, AIDS and Psychiatry

HIV and the Brain

- The virus enters the central nervous system shortly after the initial HIV infection and may cause meningitis or encephalitis.
- Viral load is more closely associated with the degree of cognitive impairment. Patients with serum viral loads $\geq 30,000$ copies/mL are 8.5 times more likely to develop dementia compared to patients with viral loads $< 3,000$ copies/mL (Childs et al., 1999). In another study, a cerebrospinal fluid viral load > 200 copies/mL was predictive of progression to neuropsychological impairment (Ellis et al., 2002).

HIV-Associated Dementia

This syndrome has been referred to by various names:

1. **HIV-associated dementia complex (HAD)** (Working Group of the American Academy of Neurology AIDS Task Force, 1991)
2. **HIV encephalopathy**
3. **Subacute encephalitis** (Snider et al., 1983)
4. **AIDS encephalopathy and AIDS-dementia complex** (Navia et al., 1986b).

HIV-associated dementia is defined as acquired cognitive abnormalities in two or more domains and is associated with functional impairment and acquired motor or behavioral abnormalities, in the absence of another etiology

The clinical manifestations of HAD are predominantly those of subcortical dementia with deficits in Cognitive, Motor and Behavioral domains.

- The cognitive disturbance usually has a subtle onset and involves slowed thinking, memory impairment, forgetfulness and difficulty concentrating. Frank disorientation, confusion and muteness are observed late in the illness. Psychiatric symptoms, such as agitation, mania, hallucinations and paranoid delusions are not uncommon in advanced stages.
- When motor signs and symptoms occur in the absence of HIV dementia, the syndrome is known as **HIV associated myelopathy**.
- **Behavioral symptoms** include social withdrawal, apathy, sleep disturbances, fatigue, headaches and decreased libido. These features may be difficult to distinguish from depressive symptoms, though the patient often lacks the dysphoria experienced in a clinical major depression.

A less severe form of CNS disease associated with HIV infection is **HIV-associated minor cognitive-motor disturbance**, which is differentiated from HAD based upon the presence of fewer symptoms and little or no functional impairment.

Opportunistic Infections and other causes of neuropsychiatric disturbance

Infection with HIV may indirectly lead to neuropsychiatric disturbances due to **CNS opportunistic infections, neoplasms and metabolic disorders**. Less common with the advent of HAART (Highly active anti-retroviral therapy). The usual HAART regimen combines three or more different drugs such as two nucleoside reverse transcriptase inhibitors (NRTIs) and a protease inhibitor (PI), two NRTIs and a non-nucleoside reverse transcriptase inhibitor (NNRTI) or other such combinations.

CNS Complications of HIV Infection

Direct effects of HIV

- Headache
- Aseptic meningitis
- HIV-associated dementia (HAD)
- Minor cognitive-motor disorder (MCMD)
- Vacuolar myelopathy

Opportunistic infections

- Cerebral toxoplasmosis
- Cryptococcal meningitis
- Progressive multifocal leukoencephalopathy (PML)
- Cytomegalovirus (CMV) encephalitis
- Tuberculous meningitis
- Neurosyphilis

Opportunistic neoplasms

- Primary CNS lymphoma (EBV-related)
- Metastatic systemic lymphoma
- Kaposi's syndrome

Source: Horwarth E and Nash SS (2005)

Neuropsychiatric Side Effects of Drugs

Drug	Side Effects
NRTIs	
Zidovudine (ZDV, AZT)	Agitation, confusion, lethargy, headache, insomnia
Didanosine (ddI)	Peripheral neuropathy
Zalcitabine (ddC)	Peripheral neuropathy
Stavudine (d4T)	Peripheral neuropathy
NNRTI	
Efavirenz	Nightmares, hallucinations, confusion, impaired concentration, somnolence, insomnia, euphoria, dizziness
PIs	
Ritonavir	Paresthesias
Saquinavir	Headache
Amprenavir	Headache, paresthesias
Other Agents	
Acyclovir	Hallucinations
Gancyclovir	Confusion, mania
Dapsone	Agitation, hallucinations
Corticosteroids	Mood liability, mania, psychosis
Sulfadiazine	Encephalopathy, hallucinations
Amphotericin B	Agitation, lethargy, disorientation, headache
Vincristine	Lethargy, dysphoria, seizures

Reference

1. By Ewald Horwath, MD and Sara Siris Nash, MD, *Psychiatric Times*, November 2005

Drug Misuse in Pregnancy

Potential difficulties faced by the children of drug users

- Physical and emotional abuse or neglect
- Inadequate parenting or supervision
- Separation
- Poverty
- Poor education
- Exposure to criminal behaviour
- Social isolation

Potential impact of parental drug use on children's welfare

- Impaired judgement, coordination and consciousness can affect a parent's ability to care for and supervise young children
- Drug-induced disinhibition can lead to aggressive behaviour, including domestic violence
- Withdrawal from certain drugs can cause irritability and mood disturbance
- Unemployment, poverty and criminality may impair family functioning
- Drug use can become a higher priority for the parent than buying basic essentials for the family
- Reduced parental vigilance may leave children vulnerable to abuse by visitors to the home
- The presence in the home of drugs and/or injecting equipment puts children at risk (National Treatment Agency for Substance Misuse, 2002; Advisory Council on the Misuse of Drugs, 2003; Department for Education and Skills, 2003)
- Disruption in family rituals, routines, communication, social life and impairment of finances.
- In many cases, children's welfare can be safeguarded by appropriate health and social care without recourse to formal child protection measures
- A child protection case conference may be convened to determine the facts and decide on further action. The child's name may be placed on the child protection register or proceedings may be instituted for a care or supervision order if the case warrants such intervention

Negative experiences of children and adolescents living with parental substance misuse

- High levels of violence
- Experiencing or witnessing neglect or abuse – physical, verbal or sexual
- Poor and/or neglectful parenting
- Inconsistency from one or both parents
- Having to adopt responsible or parenting roles at an early age

- Feeling negative emotions such as shame, guilt, fear, anger and embarrassment
- Possible neurodevelopmental consequences of substance misuse in pregnancy (e.g. foetal alcohol syndrome) that may contribute to developmental delays or intellectual disability
- Behavioural and emotional difficulties.

Effects of drug use on the child

As a general principle, exposure to substances in the first trimester of pregnancy affects **foetal organogenesis**, whereas use in the second and third trimesters mainly results in **growth and functional abnormalities or impairments in the newborn**. Persistent drug use close to term can result in preterm labour, sudden infant death syndrome (SIDS) and neonatal abstinence syndromes.

Opioids

Dependent heroin use during pregnancy is associated with a reduction of foetal growth, resulting in low birth weight, prematurity, and foetal and neonatal death (Hulse et al, 1997, 1998; Dunlop et al, 2003).

The clinical signs of opioid neonatal abstinence syndrome occur in 48–94% of infants exposed to opioids in utero, with signs of withdrawal from methadone being more common than from heroin.

Characteristics of opioid neonatal abstinence syndrome

- Gastrointestinal disturbances
- Irritability
- Hyperactivity
- Feeding and sleeping disturbances
- Autonomic hyperactivity
- Seizures (these are rare, occurring in less than 5%)

(Osborn et al, 2004)

Blood-borne viral infections

Intravenous drug users who share needles or injecting paraphernalia are at particular risk of contracting blood-borne viral infections such as HIV, hepatitis B and hepatitis C. Pregnant women who are infected can transmit the infection to their babies (vertical transmission) during pregnancy, during the process of birth or through breast-feeding.

Cocaine and Amphetamines

Foetal and neonatal difficulties associated with maternal cocaine use (and potentially amphetamine use)

Foetal difficulties associated with use during pregnancy

- Placental abruption
- Intrauterine growth retardation
- Spontaneous abortion
- Pre-eclampsia
- Pulmonary oedema
- Seizures
- Cardiac arrhythmias
- Congenital physical anomalies affecting mostly the ocular and urogenital systems

Neonatal difficulties associated with heavy use near term

- Neonatal intoxication, with symptoms including:
 - Irritability
 - hypertonia
 - sleep and appetite disturbances (Bandstra & Burkett, 1991)

Benzodiazepines

Benzodiazepines also have low teratogenic potential, although high-dosage use during pregnancy has been associated with abnormalities such as cleft lip and cleft palate. Continued maternal use near term can result in 'floppy baby syndrome', characterised by lethargy, irritability, reduced muscle tone and respiratory depression in the newborn, and a neonatal abstinence syndrome (Sanchis et al, 1991).

Longer-term effects

Maternal drug use may continue to have an impact on the child's cognitive, educational, emotional and behavioural development throughout early life.

Management Plan

- Well-integrated care package by a multi-disciplinary team in a shared specialised clinic. The specialist drug service needs to work in conjunction with the general practitioner, midwife, obstetrician, paediatrician and social worker in organising and providing care for drug-dependent women throughout the antenatal, intranatal and postnatal periods.
- The pregnant drug user (and possibly her partner) needs to be actively involved in all phases of planning and decision-making.
- Assessment of drug use should include the type and quantity of drug used, route of administration, injecting behaviour, degree of dependence, previous treatment history and motivation to change.
- The assessment process must consider the risk to the physical and mental health of the mother during pregnancy, the risk of teratogenicity or the development of a neonatal abstinence syndrome, and ongoing child care and parenting issues.

Assessment factors specific to pregnant women (in addition to a mental health history)

- Past obstetric and gynaecological history
- Screening for sexually transmitted diseases and blood-borne viral infections such as HIV and hepatitis B and C
- Feelings about the pregnancy and the baby
- Plans for the baby's future
- Feelings of guilt or self-blame
- Fears about the baby being taken into care
- Existing needs and available resources

Neonatal abstinence syndromes

Drug-using women should be prepared for the possibility that their newborn child will experience withdrawal symptoms (neonatal abstinence syndrome). Such preparation may help to allay their fears and engage them in treatment. They may be advised to stay in hospital for at least 3 days after delivery, to allow monitoring of the child for a neonatal abstinence syndrome.

Supportive measures for a neonate with a neonatal abstinence syndrome

- Keep the baby in a quiet, dimly lit room
- Ensure close, gentle interaction with the mother
- Provide small, frequent feeds
- Ensure close monitoring by nursing and medical staff

Pharmacological management

Opioid dependence

1. Methadone

- Methadone maintenance treatment during pregnancy, when combined with a comprehensive psychosocial treatment programme, can reduce the incidence of obstetric and foetal complications as well as neonatal morbidity and mortality (Finnegan, 1991; Ward et al, 1998).
- Methadone maintenance therapy should be started as soon as possible after confirmation of pregnancy. If the woman is already on a methadone programme, maintenance should be continued.
- Detoxification from all drugs is unrealistic for most of this population, and often results in the mother experiencing an abstinence syndrome leading to foetal distress (Ward et al, 1998).
- The overall aim should therefore be to maintain the woman on methadone for

the entire pregnancy, as withdrawal may lead to a risk of miscarriage in the first trimester or premature labour and foetal death in utero in the third trimester (Finnegan, 1991).

- The metabolism of methadone is increased by pregnancy, and this may cause previously stable women to experience withdrawal symptoms in the final trimester. If not carefully managed this can lead to an increased risk of relapse. An alternative to increasing the daily methadone dose is to use a split dose in order to maintain steadier plasma levels (Wittman & Segal, 1991).

2. Buprenorphine

- Buprenorphine shows considerable potential as a treatment for opioid-dependent pregnant women, and may be associated with a low incidence of neonatal abstinence syndrome (Fischer et al, 2000). **However, as yet there is insufficient controlled research with adequate follow-up periods to demonstrate its safety during pregnancy and breast-feeding**

Cocaine dependence and others

- Combination of symptomatic interventions during the withdrawal phase and psychosocial interventions. (limited evidence base)
- A similar approach should be adopted in managing the use of other psychostimulant drugs such as amphetamines and methylenedioxymethamphetamine (MDMA, ecstasy), where the evidence base is also limited.

Benzodiazepine dependence

- Sudden cessation of benzodiazepine use can lead to maternal convulsions and so should be avoided (Hepburn, 2002).
- The primary aim in treating benzodiazepine dependence in pregnant women is usually to identify a **mutually agreeable and realistic goal, be it low-dose 'maintenance', gradual reduction or detoxification.**
- For women using high doses of benzodiazepine alone, without any significant psychosocial or medical complications, **gradual reduction and detoxification in the community are recommended.**
- Women who are taking short- or medium-acting benzodiazepines (e.g. lorazepam, oxazepam) should be transferred to an equivalent dose of diazepam and the dose gradually reduced to zero.
- Women who are using high doses of benzodiazepine in combination with other drugs, or those who have complicating medical, psychiatric or psychosocial problems, are best managed in hospital.
- Once admitted for detoxification, the level of withdrawal symptoms and other problems should be objectively assessed.
- With long-acting benzodiazepines, symptoms of withdrawal may not be manifest for the first 5–7 days, and post-withdrawal problems such as sleep disturbance may take several weeks to resolve. **Pharmacological treatment is best supplemented with individual supportive psychotherapy, anxiety management and other supportive measures.**

Alcohol

- Alcohol is an established human teratogen and there is no clear safe level of consumption during pregnancy. Consumption of less than 7 units per week is thought to cause no significant harm to the baby, but more regular use can adversely affect the developing foetal brain and result in a series of physical, neurological and behavioural abnormalities known as the foetal alcohol syndrome (Jones & Smith, 1973). **It is therefore recommended that pregnant women abstain from drinking alcohol.**

Reference

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Conduct Disorder Management

(Useful for general principles in managing children with difficult behaviours)

1. Engage the family

Once engaged, the quality of the therapist's alliance with the family affects treatment success: in one meta-analysis it accounted for 15% of the variance in outcome

2. Select which treatment to use and who should deliver it

- For example, improvements in the home arising from a successful parent training programme will not necessarily lead to less antisocial behaviour at school (Scott, 2002).
- If classroom behaviour is a problem and a school visit shows that the teacher is not using effective methods, advice to the teacher and other school staff can be very effective.
- If the child has pervasive problems including fights with peers, individual work on anger management and social skills should be added.

3. Develop strengths

- Identifying the strengths of both the child and the family is crucial. This helps engagement, and increases the chances of effective treatment.
- Encouraging their abilities helps the child spend more time behaving constructively rather than destructively – more time spent playing football is less time spent hanging round the streets looking for trouble.
- Encouraging prosocial activities – for example to complete a good drawing or to play a musical instrument well – may lead to increased achievements, heightened self-esteem and greater hope for the future.

4. Treat comorbid conditions

- Depression and attention-deficit hyperactivity disorder (ADHD); a number have post-traumatic stress disorder (PTSD), for example witnessing domestic violence.

5. Promote social and scholastic learning

6. Treat the child in their natural environment

Specific interventions for children 3–12 years of age

Parent management training

Key targets of parenting skills include:

- promoting play
- developing a positive parent–child relationship
- using praise and rewards to increase desirable social behaviour
- giving clear directions and rules
- using consistent and calmly executed consequences for unwanted behavior
- reorganising the child's day to prevent problems

The **Helping the Non-compliant Child programme** (McMahon & Forehand, 2003) and **parent–child interaction therapy** (PCIT; Eyberg, 1988) are two examples of well-validated individual interventions.

The four common targets of cognitive–behavioural and social skills therapies

- To **reduce children's aggressive behaviour** such as shouting, pushing, and arguing
- To **increase prosocial interactions** such as entering a group, starting a conversation, participating in group activities, sharing, cooperating, asking questions politely, listening and negotiating
- To **correct the cognitive deficiencies**, distortions and inaccurate self-evaluation exhibited by many of these children
- To **ameliorate emotional dysregulation and self-control problems** so as to reduce emotional lability, impulsivity and explosiveness, enabling the child to be more reflective and able to consider how best to respond in provoking situations

Two of the more popular treatment models are **problem-solving skills training with in vivo practice** (PSST-P; Kazdin, 1996b) and the **Coping Power Program** (Lochman & Wells, 2002).

Interventions in school

Interventions to promote positive behavior

The four common targets of classroom techniques

- **Promoting positive behaviours** such as compliance and following established classroom rules and procedures
- **Preventing problem behaviours** such as talking at inappropriate times and fighting
- **Teaching social and emotional skills** such as conflict resolution and problem-solving
- **Preventing the escalation** of angry behaviour and acting out

Interventions for teenagers

In adolescence, conduct disorder frequently becomes more serious and can involve criminal offending.

Family-based interventions

The best known intervention for serious antisocial behaviour in teenagers is functional family therapy, brought into being in 1969 by James Alexander and colleagues (Alexander et al, 2000)

The four phases of functional family therapy

- Engagement
- Motivation
- Behavioural change
- Generalisation

The principles of multisystemic therapy

- An assessment is made to **determine the fit between the young person's problems and the wider environment**, identifying **strengths and difficulties**; difficulties are understood as reactions to a specific context, not necessarily as intrinsic deficits
- During sessions the **therapist emphasises the positive and uses systemic strengths** (e.g. an aptitude for sports, getting on well with grandmother, the presence of prosocial peers in grandmother's neighbourhood) as levers for change. Each session should acknowledge and work on these strengths
- Interventions are designed to **promote responsible behaviour and reduce irresponsible behaviour**
- Interventions are **focused in the present and are action oriented, with specific, well-defined goals**. The emphasis is on what can be done in the here and now, rather than on the need to understand the family and the youth's past
- Interventions target sequences of behaviour in multiple systems that maintain problems
- Interventions are **developmentally appropriate**. They should fit the life stage and personal level of the family members
- Interventions **require daily or weekly effort by family members (implementation)**. This enables frequent practice of new skills, frequent positive feedback for efforts made; non-adherence to treatment agreements rapidly becomes apparent
- The **effectiveness of interventions is evaluated continuously from multiple perspectives (evaluation)**, with the multisystemic therapy team assuming responsibility for overcoming barriers to successful outcomes.
- Interventions are designed to promote **treatment generalisation** by empowering parents to address their offspring's needs across multiple contexts

Medication (Adolescents with conduct disorder)

- Psychostimulants (methylphenidate and dexamphetamine), used for ADHD comorbid with conduct disorder.
- Mood stabilisers (e.g. lithium and carbamazepine) and drugs purported to target affect dysregulation (e.g. buspirone and clonidine).
- Lithium reduced aggression and hostility in children and adolescents in psychiatric hospitals
- The review by Pappadopulos et al(2006) found that antipsychotics were more effective where ADHD or intellectual disability was present.

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Assessment and Management of Insomnia

Insomnia is an experience of inadequate or poor-quality sleep characterized by one or more of the following problems: **difficulty falling asleep, difficulty maintaining sleep, waking up too early in the morning and sleep that is not refreshing**. Insomnia also involves daytime consequences such as fatigue, lack of energy, difficulty concentrating and irritability.

Insomnia disorder in the DSM V

- Predominant complaint of dissatisfaction with sleep quantity or quality, associated with one (or more) of the following symptoms
 1. Difficulty initiating sleep
 2. Difficulty maintaining sleep, characterized by frequent awakenings or problems returning to sleep after awakenings
 3. Early morning awakening with inability to return to sleep
- The sleep disturbance causes clinically significant distress or impairment in social, occupational, educational, academic, behavioral or other important areas of functioning.
- The sleep difficulty occurs at least 3 nights per week and is present for at least 3 months.

Types of Insomnia

1. Acute Insomnia

Acute insomnia is often caused by an **emotional or physical discomfort**. Some examples of common discomforts include significant life stress, acute illness and environmental disturbances such as noise, light and temperature. Sleeping at a time that is inconsistent with the daily biologic rhythm, such as occurs in persons with jet lag, also may cause acute insomnia.

2. Chronic Insomnia

Chronic insomnia may be caused by many different factors acting singly or in combination and often occurs in conjunction with other health problems.

Insomnia and Psychiatric, Medical and Neurologic Disorders.

- **Psychiatric Disorders: Mood and anxiety disorders are the most common psychiatric diagnoses associated with insomnia.**

Medical disorders

- Chronic Pain Syndromes
- Menopause

- Gastroesophageal Reflux and Peptic Ulcer Disease
- COPD/Asthma
- Benign Prostatic Hyperplasia
- Prescription and nonprescription drugs, and drugs of abuse (e.g., heroin, cocaine, other stimulants) cause increased wakefulness and poor-quality sleep.
- Stimulating antidepressants, steroids, decongestants, beta blockers, caffeine, alcohol, nicotine and illicit drugs.
- *Insomnia and Specific Sleep Disorders.*

Insomnia may be associated with specific sleep disorders, including restless legs syndrome, periodic limb movement disorder, sleep apnea and circadian rhythm sleep disorders.

- **Restless Legs Syndrome.** Restless legs syndrome is characterized by unpleasant sensations in the legs or feet that are temporarily relieved by moving the limbs. Symptoms increase in the evening, especially when a person is lying down and remaining still. The dysesthesias cause difficulty falling asleep and are often accompanied by periodic limb movements.
- **Periodic Limb Movement Disorder.** Periodic limb movement disorder is characterized by bilateral repeated, rhythmic, small-amplitude jerking or twitching movements in the lower extremities and, less frequently, in the arms. These movements occur every 20 to 90 seconds and can lead to arousals, which are usually not perceived by the patient. Rather, the patient reports that sleep is not refreshing. Characteristically, the bed partner is more likely to report the movement problem. This condition and restless legs syndrome are more common in older patients.
- **Obstructive Sleep Apnea.** Obstructive sleep apnea is most commonly associated with snoring, daytime sleepiness and obesity but occasionally presents with insomnia.
- **Circadian Rhythm Sleep Disorders.** Circadian rhythm sleep disorders are characterized by an inability to sleep because of a mismatch between the circadian sleep rhythm and the desired or required sleep schedule. Two subtypes

Delayed sleep phase syndrome

- Difficulty falling asleep at the desired time
- Difficulty waking at the desired time

Advanced sleep phase syndrome

- Difficulty staying awake in the evening
- Waking too early

Assessment of Insomnia

- How has the patient been sleeping recently?
- When did the problem begin? (To differentiate between acute and chronic insomnia)

- Does the patient have a psychiatric or medical condition that may cause insomnia? (May relate to an underlying condition that should be treated first)
- Is the sleep environment conducive to sleep? (Noise, interruptions, temperature, light)
- Does the patient report "creeping, crawling or uncomfortable feelings" in the legs that are relieved by moving the legs? (May relate to restless legs syndrome)
- Does the bed partner report that the patient's legs or arms jerk during sleep? (May relate to periodic limb movements in sleep)
- Does the patient snore loudly, gasp, choke or stop breathing during sleep? (May relate to obstructive sleep apnea)
- Is the patient a shift worker? What are the work hours? Is the patient an adolescent? (May relate to circadian sleep disorders/sleep deprivation)
- What are the bedtimes and rise times on weekdays and weekends? (May relate to poor sleep hygiene)
- Does the patient use caffeine, tobacco or alcohol? Does the patient take over-the-counter or prescription medications (such as stimulating antidepressants, steroids, decongestants, beta blockers)? (May relate to substance-induced insomnia)
- What daytime consequences does the patient report? (Daytime consequences may be significant)
- Does the patient report dozing off or having difficulty staying awake during routine tasks, especially while driving? (This is a serious problem that should be dealt with promptly)

Sleep Diary

Sleep diary for one to two weeks. Sleep diaries usually record bedtime, total sleep time, time until sleep onset, number of awakenings, use of sleep medications, time out of bed in the morning and a rating of quality of sleep and daytime symptoms. The sleep diary provides a night-to-night account of the patient's sleep schedule and perception of sleep. Moreover, it may serve as a baseline for assessment of treatment effects.

Important Points in Assessment

- Major Depressive Episode
- Generalized Anxiety or Panic Disorder
- Excessive daytime sleepiness (unexpected or irresistible sleepiness) resulting in imminent risk to the patient and/or society
- Substance abuse

Management of Insomnia

When the insomnia persists beyond one or two nights or becomes predictable, treatment should be considered. Pharmacologic treatment usually predominates, especially short-acting hypnotics.

1. General Sleep Hygiene Measures

- Wake up at the same time each day.
- Discontinue caffeine intake four to six hours before bedtime and minimize total daily use. Caffeine is a stimulant and may disrupt sleep.
- Avoid nicotine, especially near bedtime and on night awakenings. It is also a stimulant.
- Avoid the use of alcohol in the late evening to facilitate sleep onset. Alcohol can cause awakening later in the night.
- Avoid heavy meals too close to bedtime, since this may interfere with sleep. A light snack may be sleep inducing.
- Regular exercise in the late afternoon may deepen sleep. Vigorous exercise within three to four hours of bedtime may interfere with sleep.
- Minimize noise, light and excessive temperatures during the sleep period.
- Move the alarm clock away from the bed if it is a source of distraction.
- Avoid using I-pad, I-phone or other mobile devices as reading tools.
- Since chronic insomnia is often multifactorial in etiology, a patient may need multiple treatment modalities. **If an underlying medical or psychiatric condition is identified, this condition should be treated first.**
- If chronic insomnia appears to be primary or persists after treatment of an underlying condition, two general treatment approaches are available: behavior and pharmacologic

2. Behaviour Therapy

- **Relaxation techniques**
- **Sleep Restriction Therapy:** Sleep restriction therapy consists of curtailing the amount of time spent in bed to increase the percentage of time spent asleep. This improves the patient's sleep efficiency (time asleep/time in bed).
- **Stimulus Control Therapy.** Stimulus control therapy is based on the premise that insomnia is a conditioned response to temporal (bedtime) and environmental (bed/bedroom) cues usually associated with sleep. **The main objective of stimulus control therapy is to reassociate the bed and bedroom with the rapid onset of sleep.** Stimulus control instructions involve **going to bed only when sleepy; using the bed and bedroom only for sleep; getting out of bed and going into another room when unable to fall asleep or return to sleep easily and returning to bed only when sleepy again; maintaining a regular morning rising time regardless of duration of sleep the previous night; and avoiding daytime naps.**

3. Cognitive Therapy. Cognitive therapy involves **identifying dysfunctional beliefs and attitudes about sleep and replacing them with more adaptive substitutes.** For example, patients who believe that sleeping eight hours per night is absolutely necessary to function during the day are asked to question the evidence and their own experience to see if this is true for them.

4. Pharmacological Treatment

- Hypnotic Medications. Benzodiazepines and Z-Drugs (Zopiclone, Zolpidem)
- **Antidepressants:** Sedating antidepressants particularly if depression is present. E.g mirtazapine, amitriptyline
- **Antihistamines**
- **Melatonin:** Melatonin has several physiologic actions, including a phase-shifting effect on circadian rhythms, increased sleepiness when administered during daytime hours and vasoconstriction. Its mechanisms of action are unknown but may involve interaction with melatonin receptors in the suprachiasmatic nucleus.
- Finally, a variety of herbal preparations (e.g., valerian root, herbal teas), so-called nutritional substances (e.g., l-tryptophan) and over-the-counter drugs are promoted.
- In the elderly, be aware of medical issues, risk of falls with hypnotics and pharmacokinetic considerations.

Reference

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